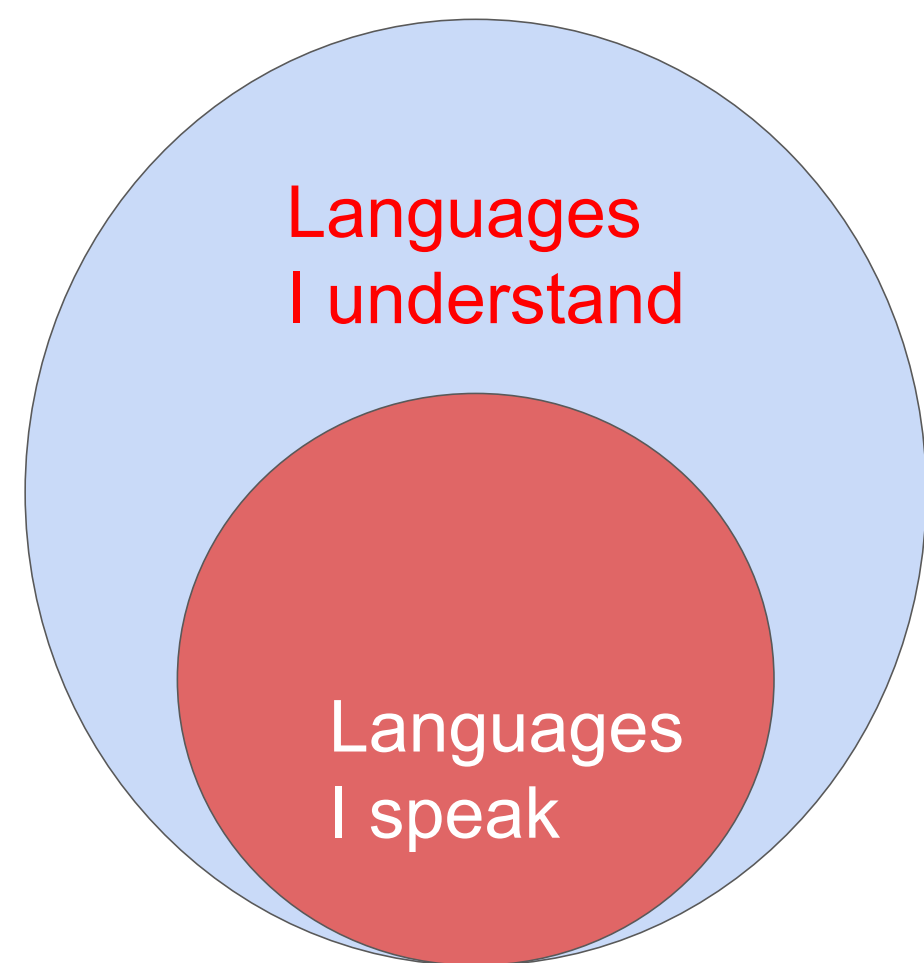


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What, Why and How?



Receptive Multilingualism

- Slavic languages are typologically similar and make good candidates for examining "Receptive Multilingualism".
- Can speakers use their L2 to understand Lx due to factors such as orthographic, phonological, and lexical similarities between Slavic languages?
- Ukrainian (UK) = Belarusian (BE) phonology, morphology + Czech (CS) phonology + Polish (PL) morphology, lexicon + Bulgarian (BL) and Russian (RU) lexicon → optimal Slavic bridge language

Experiment and Data Analysis

- Data from [1]. Two groups of participants (with and without UK knowledge) translating BE, BL, CS and PL into RU in a web-based experiment.
- Translation tasks across Spoken, Written and mixed modalities:
Example: "Give me at least some kind of bonus or something?"
 - Free translation - Open text response
 - MCQ translation - Choose best of 2 optionsOptions:
 - "Or what?"
 - "Or anything?"
- Predictor variables for the analysis:
 - Levenshtein Distance
 - Jaccard Similarity
 - LLM Surprisal of the phrase and the sentence
- Response variables for the analysis:
 - Accuracy (separate for each task)
 - COMET [2]
 - BERTScore (F1-Score) [3] } Translation Quality



Fig 1. Free Translation Task from [1]

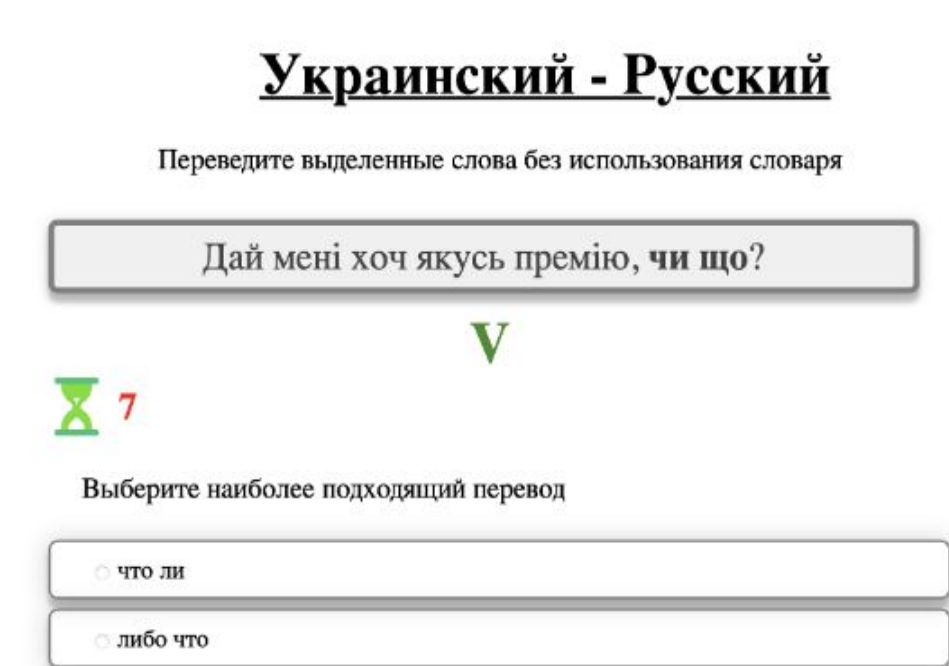
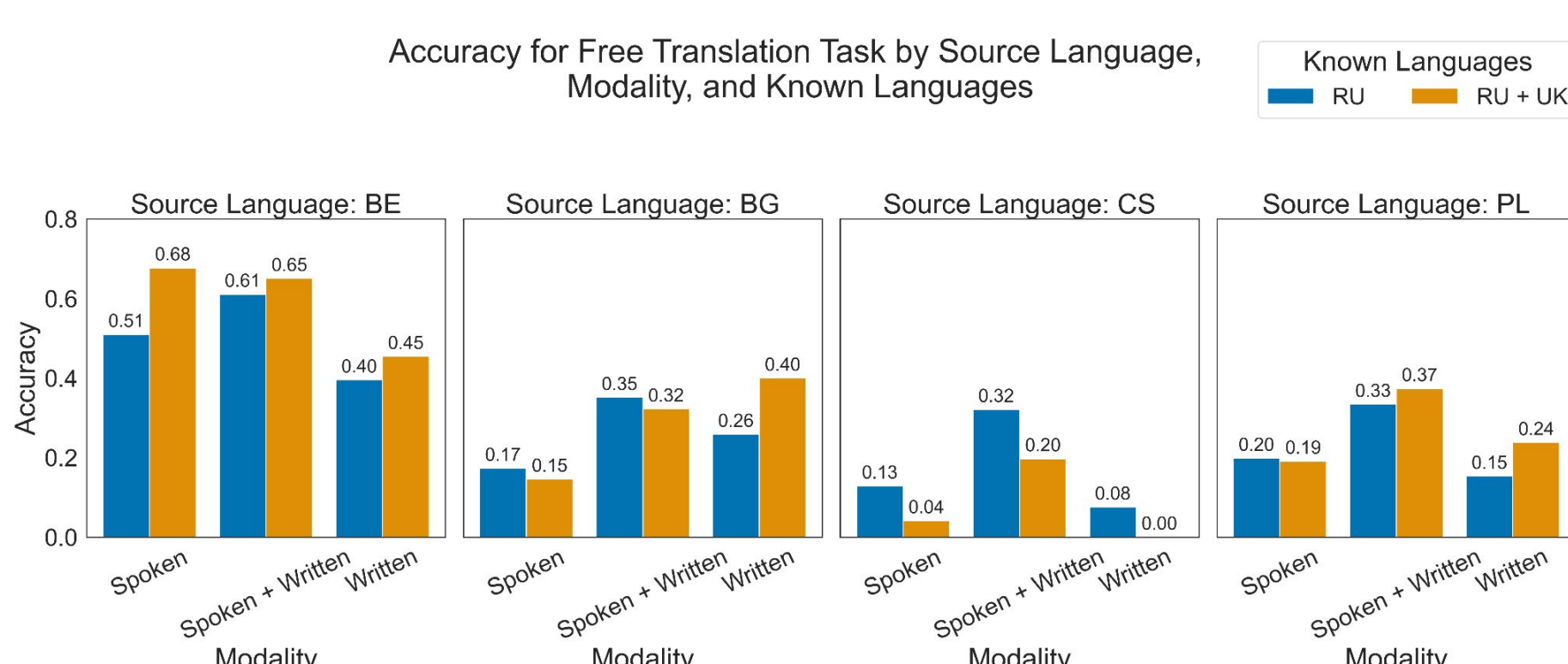
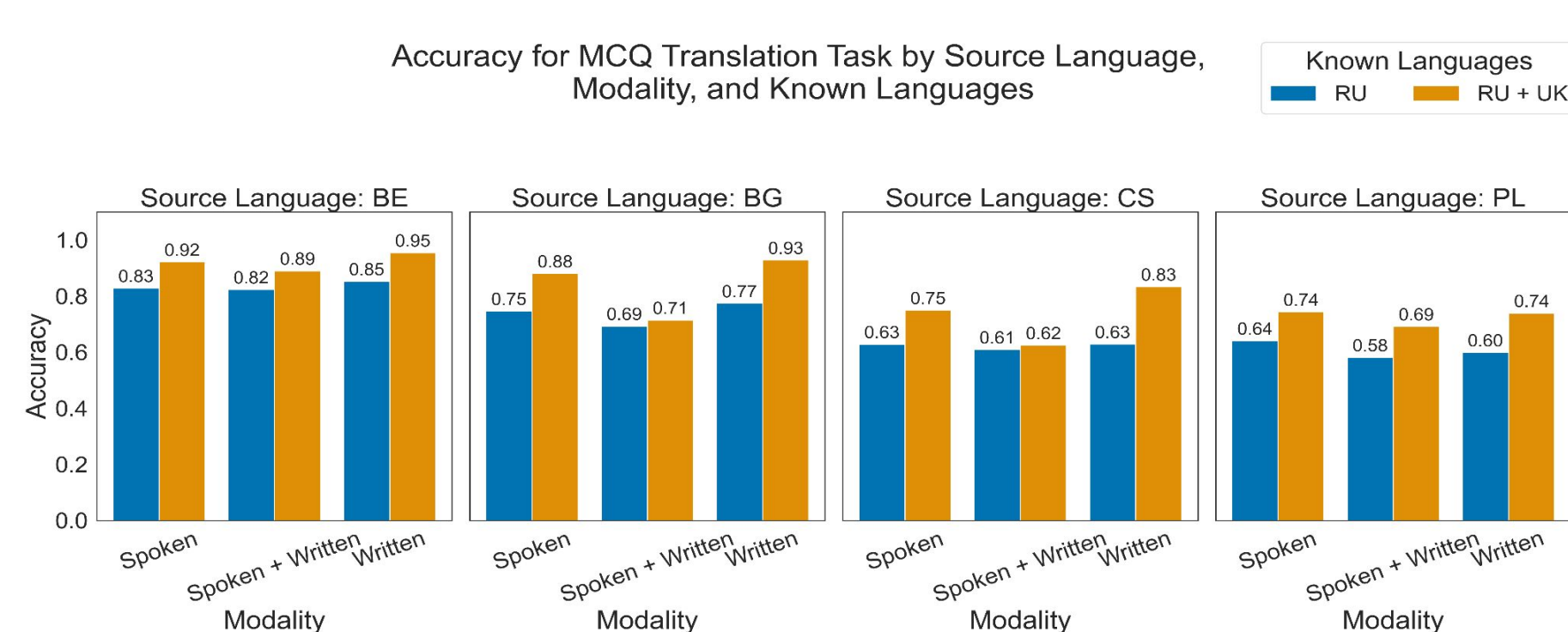


Fig 2. MCQ Translation Task from [1]

RQ1: How well do native Russian speakers with Ukrainian as L2 understand unfamiliar Slavic languages (Lx) compared to monolingual Russian speakers?

- **233 RU speakers** and **20 RU+UK speakers** in the MCQ task.
- **206 RU speakers** and **20 RU+UK speakers** in the Free translation task.
- Participants with Ukrainian knowledge tend to perform worse when translating CS during Free translation.
- Participants tend to perform best when BE is the source language.
- Higher sentence surprisal might indicate that participants are likely to engage in more careful analysis.



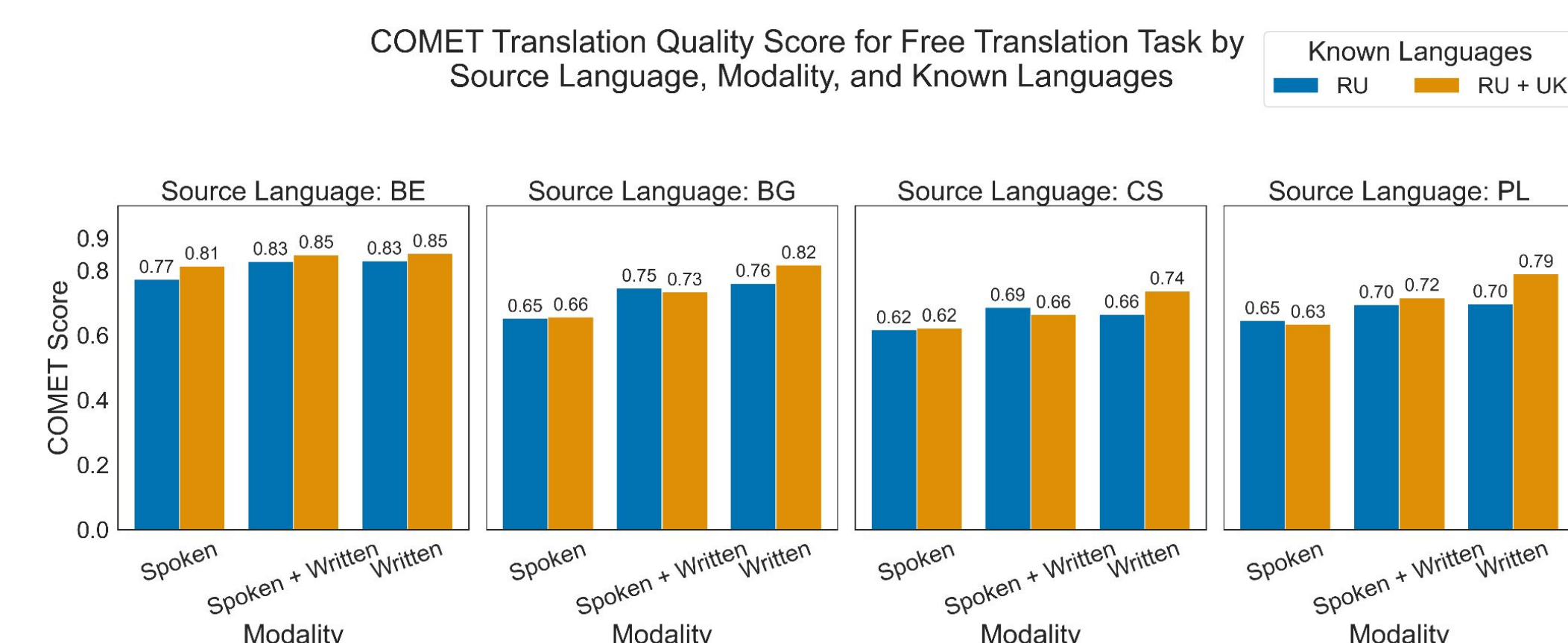
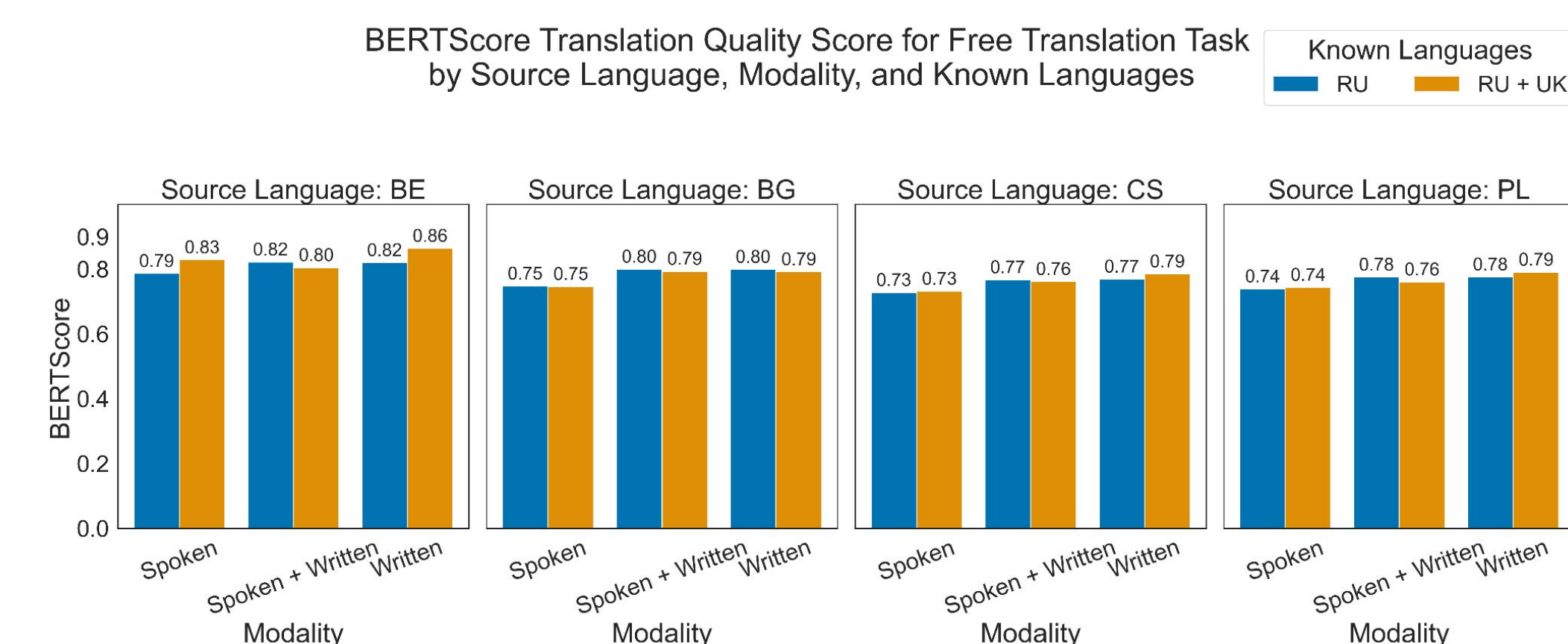
How Variables Affect Translation Accuracy According to Models (tested in R)

Variables	Free Translation Accuracy	MCQ Translation Accuracy
RU+UK speakers	-	↑
Source language with BE as baseline	↑	↑
Lower Levenshtein distance	↑	-
Higher Jaccard similarity	↑	↑
Interaction between RU+UK speakers and BE as a source language	↑	-
Interaction between the written modality and BG as a source language	↓	↑
Higher source phrase surprisal	↓	↓
Higher sentence level surprisal	-	↑

TL;DR

- Slavic languages were used to examine receptive multilingualism in spoken, written, and mixed modalities due to shared typological features.
- Knowledge of Ukrainian generally helps with understanding other Slavic languages, with the exception of Czech in Free translation.
- Belarusian as source language overall with best translation accuracy and quality across speakers.
- Imbalance of monolingual vs. bilingual speakers lowers power of statistical analysis: need a more balanced dataset for future analysis.

RQ2: Do these two groups differ in their translation quality, as evidenced by COMET and BERTScore?



- Participants tend to perform worse in the spoken modality.
- Performance varies by source language.
- Participants tend to perform best when BE is the source language.
- Less deviation of score in the BERTScore metric.

How Variables Affect Scores According to Models (tested in R)

Variables	COMET Score	F1-Score (from BERTScore)
RU+UK speakers	↑	↑
Lower Levenshtein distance	↑	↑
Higher Jaccard similarity	↑	↑
Source languages other than BE	↓	↓
Interaction between RU+UK speakers and BG as a source language	↓	↓
Interaction between the spoken modality and BG as a source language	↓	↓
Higher source phrase surprisal	↑	↓

Conclusion

- In MCQ task, Russian speakers with Ukrainian as L2 performed better than Russian monolinguals in all four Slavic languages to varied degrees.
- Russian speakers with Ukrainian L2 in the free translation task performed worse than Russian monolinguals in Czech.
- Russian speakers with Ukrainian knowledge also had better translation quality, as indexed by COMET and BERTScore.
- Ukrainian can serve as a bridge language for Russian speakers to understand a foreign Slavic Lx, but to different degrees depending on the task, the modality of the task, and the typological similarity of the foreign Slavic Lx to Russian and Ukrainian.
- Some limitations include:
 - Imbalance in the class size leading to lower power.
 - All translations and text to be translated were transliterated into Cyrillic and this process was lossy (CS and PL diacritics were lost).

References

- [1] I. Zaitova, I. Stenger, M. Butt, and T. Avgustinova. 2024. Cross-linguistic processing of non-compositional expressions in Slavic languages. In *Proceedings of the Workshop on Cognitive Aspects of the Lexicon @ LREC-COLING 2024*, pp. 86–97, Torino, Italia.
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- [3] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi. 2020. BERTScore: Evaluating text generation with BERT.

